

Model 1124 communications satellite

Specifications and assembly instructions



Description

The model 1124 satellite is an HG-5 based relay satellite that takes advantage of the wide angle of the HG-5 antenna to cover a large footprint on planetary surfaces. With 4 antennas the satellite to relay to many targets and can be used as part as a mesh network.

Variants

The satellite variants can be identified by the following suffixes

Model 1124[Y]

Where Y defines the attachment point and can be

Suffix	Type	Notes
(no suffix)	No attachment point	
K	KAS JS-1 socket	KAS only
C	Clamp-o-Tron jr	

Uses

STC satellite

General relay

Costs and tech

Costs

Base model	9510	Funds
Model K	10260	Funds
Model C	10310	Funds

Tech requirements

Technology	Use	Notes
Electrics	Probodobodyne OKTO Z-200 Battery OX-STAT Panel	
Basic science	Z-100 Battery HG-5 Antenna	
Engineering 101	Communotron 16	
Propulsion systems	LV-1 Ant engine OSCAR-B tank	
Miniaturization	Clamp-O-Tron jr.	Model C only

Advanced construction

JS-1 socket

Model K only

Specifications

General specifications

Parameter	Value	Units	Notes
Electrical storage	3210	EC	
Electrical storage margin	8	%	kerbostationary orbit
	270	EC	kerbostationary orbit
Max orbital alt (circular)	3525	Km	limited by battery storage
Min orbital alt (circular)	2600	Km	limited by battery charger
Power consumption (absolute max)	2.51	EC/sec	
Power consumption (deployment)	0.28	EC/sec	Omnis on, HG5 off, OKTO on, wheels idle
Power consumption (mission)	2.46	EC/sec	OKTO hibernating, wheels off, radios full on
Power generation (max)	2.89	EC/sec	Assumes 0.4 EC for 100% exposure
	7.23	Exposure	
Power generation (min)	2.64	EC/sec	assumes panel ring oriented towards sun
	6.59	Exposure	assumes panel ring oriented towards sun
Power generation (avg)	2.80	EC/sec	assumes panel ring oriented towards sun
	7.00	Exposure	assumes panel ring oriented towards sun
Thrust	2	kN	
dV budget (vacuum)	1421	m/s	1122C
	1441	m/s	1122K
	1455	m/s	1122
Mass (wet)	1150	Kg	1122C
	1138	Kg	1122K
	1130	Kg	1122
Mass (dry)	750	Kg	1122C
	738	Kg	1122K

Parameter	Value	Units	Notes
	730	Kg	1122
Omni Antenna range	1875	Km	assumes root model, 0.5 range multiplier and 0.5 multiple antenna bonus

Power budget

All consumption in EC/second.

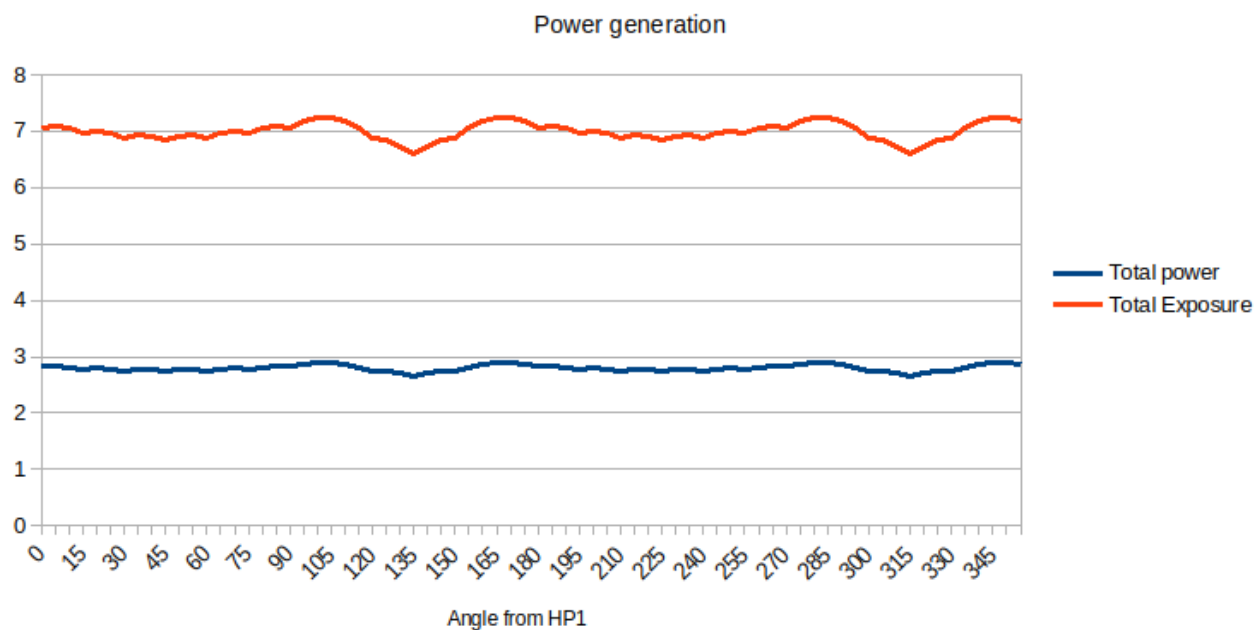
System	Unit consumption	Amount	Total consumption	Notes
Communotron 16	0.13	2	0.26	
HG-5	0.55	4	2.20	
Mission subtotal			2.46	
OKTO	0.02	1	0.02	
Systems subtotal			2.48	
Reaction wheel	0.03	1	0.03	
Abs. maximum			2.51	

Electrical generator performance

All figures assume side of power rings pointing towards the sun. Derate by cosine of angle to perpendicular of panels in other cases. Mission attitude is satellite perpendicular to ecliptic (direction measured as a line intersecting centre of computer section and centre of propulsion section).

Zero angle is at hardpoint 1 (where the computer label is).

Model 1124 communications satellite



Angle	Total power	Total Exposure	UPSR	MPSR	LPSR	T2R
0	2.83	7.07	2	1.93	1.73	1.41
5	2.84	7.09	1.99	1.88	1.81	1.41
10	2.82	7.05	1.97	1.81	1.88	1.39
15	2.78	6.96	1.93	1.73	1.93	1.37
20	2.8	6.99	1.88	1.81	1.97	1.33
25	2.78	6.96	1.81	1.88	1.99	1.28
30	2.75	6.88	1.73	1.93	2	1.22
35	2.77	6.93	1.81	1.97	1.99	1.16
40	2.77	6.92	1.88	1.99	1.97	1.08
45	2.74	6.86	1.93	2	1.93	1
50	2.77	6.92	1.97	1.99	1.88	1.08
55	2.77	6.93	1.99	1.97	1.81	1.16
60	2.75	6.88	2	1.93	1.73	1.22
65	2.78	6.96	1.99	1.88	1.81	1.28
70	2.8	6.99	1.97	1.81	1.88	1.33
75	2.78	6.96	1.93	1.73	1.93	1.37
80	2.82	7.05	1.88	1.81	1.97	1.39
85	2.84	7.09	1.81	1.88	1.99	1.41
90	2.83	7.07	1.73	1.93	2	1.41
95	2.87	7.18	1.81	1.97	1.99	1.41
100	2.89	7.23	1.88	1.99	1.97	1.39
105	2.89	7.23	1.93	2	1.93	1.37
110	2.87	7.17	1.97	1.99	1.88	1.33
115	2.82	7.05	1.99	1.97	1.81	1.28
120	2.75	6.88	2	1.93	1.73	1.22
125	2.74	6.84	1.99	1.88	1.81	1.16
130	2.7	6.74	1.97	1.81	1.88	1.08
135	2.64	6.59	1.93	1.73	1.93	1
140	2.7	6.74	1.88	1.81	1.97	1.08
145	2.74	6.84	1.81	1.88	1.99	1.16
150	2.75	6.88	1.73	1.93	2	1.22
155	2.82	7.05	1.81	1.97	1.99	1.28
160	2.87	7.17	1.88	1.99	1.97	1.33
165	2.89	7.23	1.93	2	1.93	1.37
170	2.89	7.23	1.97	1.99	1.88	1.39
175	2.87	7.18	1.99	1.97	1.81	1.41
180	2.83	7.07	2	1.93	1.73	1.41
185	2.84	7.09	1.99	1.88	1.81	1.41
190	2.82	7.05	1.97	1.81	1.88	1.39
195	2.78	6.96	1.93	1.73	1.93	1.37
200	2.8	6.99	1.88	1.81	1.97	1.33
205	2.78	6.96	1.81	1.88	1.99	1.28
210	2.75	6.88	1.73	1.93	2	1.22
215	2.77	6.93	1.81	1.97	1.99	1.16
220	2.77	6.92	1.88	1.99	1.97	1.08
225	2.74	6.86	1.93	2	1.93	1
230	2.77	6.92	1.97	1.99	1.88	1.08
235	2.77	6.93	1.99	1.97	1.81	1.16
240	2.75	6.88	2	1.93	1.73	1.22

245	2.78	6.96	1.99	1.88	1.81	1.28
250	2.8	6.99	1.97	1.81	1.88	1.33
255	2.78	6.96	1.93	1.73	1.93	1.37
260	2.82	7.05	1.88	1.81	1.97	1.39
265	2.84	7.09	1.81	1.88	1.99	1.41
270	2.83	7.07	1.73	1.93	2	1.41
275	2.87	7.18	1.81	1.97	1.99	1.41
280	2.89	7.23	1.88	1.99	1.97	1.39
285	2.89	7.23	1.93	2	1.93	1.37
290	2.87	7.17	1.97	1.99	1.88	1.33
295	2.82	7.05	1.99	1.97	1.81	1.28
300	2.75	6.88	2	1.93	1.73	1.22
305	2.74	6.84	1.99	1.88	1.81	1.16
310	2.7	6.74	1.97	1.81	1.88	1.08
315	2.64	6.59	1.93	1.73	1.93	1
320	2.7	6.74	1.88	1.81	1.97	1.08
325	2.74	6.84	1.81	1.88	1.99	1.16
330	2.75	6.88	1.73	1.93	2	1.22
335	2.82	7.05	1.81	1.97	1.99	1.28
340	2.87	7.17	1.88	1.99	1.97	1.33
345	2.89	7.23	1.93	2	1.93	1.37
350	2.89	7.23	1.97	1.99	1.88	1.39
355	2.87	7.18	1.99	1.97	1.81	1.41

Assembly instructions

The satellite is based on the SpaceBus 1 satellite design and can be built using the standard parts. The following table presents the modules stacked in order, with angle of rotation relative to the computer module.

Module	Type	Angle	Notes
Computer	Computer	0	Module defines zero angle
UPS	Type 1 PSU	0	
T2	Type 2 PSU	0	HG-5 rest on top of OX-STAT
MPS	Type 1 PSU	15	
LPS	Type 1 PSU	30	
Propulsion	Type 2 propulsion	0	

Hardpoint	Device	Notes
1	Empty	
2	Empty	
3	HG-5 antenna	Use 4x symmetry. Apply on HP 3, 4, 5 or 6 .
4	HG-5 antenna	Use 4x symmetry. Apply on HP 3, 4, 5 or 6 .
5	HG-5 antenna	Use 4x symmetry. Apply on HP 3, 4, 5 or 6 .
6	HG-5 antenna	Use 4x symmetry. Apply on HP 3, 4, 5 or 6 .

Apply 8x Z-100 batteries on propulsion unit hardpoints closest to engine, light facing towards exhaust. Apply another 8x Z-100 batteries in upper hardpoints, back to back to lower cells.